MIL-M-45147F (MU)
3 August 1973
SUPERSEDING
MIL-M-45147E (MU)
28 September 1966

MILITARY SPECIFICATION

MINE, AT, NM, M19
PARTS FOR

1. SCOPE

1.1 This specification covers parts for one type of mine designated as Mine, Anti-tank, Nonmetallic, M19.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 -	Ammunition and Special Weapons, General Specification for.

STANDARDS

MILITARY

MIL-STD-105 -	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 -	Quality Assurance Terms and Definitions.
MIL-STD-1168-	Lot Numbering of Ammunition
MIL-STD-1169-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1345

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DRAWINGS

U.S. ARMY MUNITIONS COMMAND

9209330 - Mine, AT, NM, M19, Plastic Parts Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.1.1 Reclaimed material.-A blend of reclaimed and virgin plastic material may be used provided the requirements of this specification are met. The blend shall be limited to 70 percent minimum virgin material with the exception that any proportion of reclaimed plastic material may be used in the polyethylene handles. The reclaimed material shall be clean scrap, free from any type of foreign material not related to the blend. Floor sweeping shall not be accepted as clean scrap.

3.2 Assembly.-The parts shall comply with all the requirements specified on Drawing (dwg.) 9209330 and with all requirements specified in applicable specifications.

3.3 Internal air pressure.-The mine assembly shall withstand an internal air pressure of 2 pounds, minimum (min.) per square inch gage (p.s.i.g.) with no evidence of leakage when tested as specified.

3.4 Lug pull.-The carrying cord handle lugs shall withstand a 100 pound, min. pull applied at a uniform rate not to exceed 0.5 inches per minute when tested as specified.

3.5 Susceptibility to cracking.-The mine parts assembly shall not crack when subjected to temperature conditioning at minus 65 degrees Fahrenheit plus or minus 5 degrees, for a period of 24 hours plus or minus 1/2 hour as evidenced by failure to comply with 3.3.

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3.6 Workmanship.-All parts shall be fabricated in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, porosity, warpage, excess flash, checks, chipped edges, blisters, scratches, dirt, grease, rust, corrosion products, and other foreign matter and defects which would affect thier serviceability. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

3.7 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted.

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by

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Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agents authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission.-Prior to the start of regular production the contractor shall submit a first article sample (see 6.1c) to a Government approved facility as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 50 complete mine assemblies and 50 sets of parts and sub-assemblies (i.e. 50 each of every component and sub-assembly, including at least three (3) items from each cavity used to produce plastic parts and be so identified as to permit tracing each item to the cavity in which it was molded) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials including packaging and packing shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including a certificate of conformance for materials. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

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4.2.2 First article inspection provisions

4.2.2.1 Component parts and subassemblies

4.2.2.1.1 Submission.-Fifty (50) complete sets of parts (i.e. 50 each of every component part and every sub-assembly) including at least three (3) items from each cavity used to produce plastic parts, and be so identified as to permit tracing each item to the cavity in which it was molded.

4.2.2.1.2 Inspections to be performed.-The parts and sub-assemblies will be subjected by the Government to any or all of the applicable examinations specified in 4.3.5 of this specification, and any or all of the requirements of the applicable drawings.

4.2.2.1.3 Rejection.-If any component or subassembly fails to comply with any of the applicable requirements the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a component or sub-assembly to comply with any of the stated requirements.

4.2.2.2 Mine assemblies

4.2.2.2.1 Submission.-Fifty (50) complete mine assemblies packaged and sealed as required.

4.2.2.2.2 Inspections to be performed.-The complete mine assemblies will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 per Table I below of this specification, and any or all requirements of the applicable drawings.

TABLE I

(First Article Testing Requirements)

Test Paragraph	Sample Size	Criteria
4.3.3.1	50	0 failures
4.3.3.2	32 of those tested in 4.3.3.1	0 failures
4.3.3.3	18 of those remaining in 4.3.3.1	0 failures

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4.2.2.2.3 Rejection.-If any complete mine assembly fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a mine assembly to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major

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defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraphs of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Assembly (see dwg. 9209330).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Failure to accept simulated fuze (functional)	Gage	01001
Minor:	AQL 1.50 percent		
201.	Activator wells not covered	Visual	01002
202.	Handle or carrying cord missing or insecure	Visual	01003
203.	Evidence of poor workmanship (see 3.6)	Visual	01004

4.3.2.2 Carton, sealed (see Section 5).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Contents loose upon shaking carton	Manual	02001
102.	Carton deformed or damaged so that contents are exposed or liable to become exposed	Visual	02002
103.	Sealing strip incomplete, badly wrinkled, or not in contact with surface for at least half inch around entire carton	Visual	02003
Minor:	AQL 0.65 percent		
201.	Marking missing, misleading or illegible	Visual	02004

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4.3.3 Testing

4.3.3.1 Internal air pressure (see 3.3) - Critical defect - Code No. 01005.-This test shall be performed 100 percent. The test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4. Any assembly which fails to comply with the specified requirements shall be classed defective and be removed from the lot.

4.3.3.2 Lug pull (see 3.4) - Major defect - Code No. 01006.- Sampling plans for this test shall be in accordance with Standard MIL-STD-105 using inspection level II and an AQL of 1.5 percent. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4. (Non-Destructive)

4.3.3.3 Susceptibility to cracking (see 3.5) - Major defect - Code No. 01007.-Samples of 18 mines shall be selected from each lot for these tests. The tests shall be performed as specified in 4.4.3. If one or more assemblies fail to comply with the specified requirements, the lot shall be rejected.

4.3.4 Inspection equipment.-Inspection equipment required to perform the inspection prescribed in this specification is identified in the Examination (4.3.2) and Test Method (4.4) paragraphs herein. See 6.3 for details concerning responsibilities for inspection equipment design and approval.

4.3.5 Dimensional control of (molded) parts.-In place of the normal sampling and inspections associated with the Classification of Defects, and after a curing time for the parts has been approved (see Note 2), a sample of three (3) parts (as molded) from each cavity shall be fully inspected dimensionally to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in paragraphs 4.3.5.1 through 4.3.5.7 after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. Of the three (3) samples, one (1) sample shall be the last part producted. If any defective parts are found during qualification of the cavity, the cavity producing the defective

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part will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each separate type of defect according to MIL-STD-105, using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. All cavities removed from production because of some fault, may after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the contracting officer to substantiate the request. Contractor design of gages and test equipment required to perform the inspections listed herein shall be forwarded to Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-A-M for approval prior to manufacture of equipment. The following subparagraphs identify those items and minimum inspections subject to the requirements of this paragraph.

NOTE 2.-In establishing a curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60, etc. mins.) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to Picatinny Arsenal, Dover, New Jersey, 07801 ATTN: SARPA-QA-A-M. If any new change in material, cycle time, a reworked cavity, or a new cavity is used, a new curing time shall be established and approved.

4.3.5.1 Body (see dwg. 9209331 covering a detail of dwg. 9209330).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:			
101.	Distance between base mating surfaces (fit with base).....	Gage	03001
102.	Diameter of fuze well mating surface (fit with fuze well).....	Gage	03002
103.	Diameter of filling hole.....	Gage	03003
104.	Diameter of filling hole counterbore.....	Gage	03004

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105.	Depth of filling hole counterbore	Gage	03005
106.	Diameter of activator hole (fit with activator well)	Gage	03006
107.	Diameter of activator hole counterbore	Gage	03007
108.	Depth of activator hole counterbore	Gage	03008
109.	Total height	Gage	03009
110.	Wall thickness	Gage	03010
Minor:			
201.	Inside diameter of lugs	Gage	03011
202.	Length of lugs	Gage	03012
203.	Marking misleading or unidentifiable	Visual	03013
204.	Evidence of poor workmanship (see 3.6)	Visual	03014

4.3.5.2 Base (see dwg. 9209332 covering a detail of dwg. 9209330).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:			
101.	Radii of body mating corners (fit with body)	Gage	04001
102.	Total height	Gage	04002
103.	Diameter of fuze well mating surface	Gage	04003
104.	Edge thickness	Gage	04004
105.	Diameter of activator hole (fit with activator)	Gage	04005
106.	Diameter of activator hole counterbore	Gage	04006
107.	Depth of activator hole counterbore	Gage	04007
Minor:			
201.	Ribs missing	Visual	04008
202.	Evidence of poor workmanship (see 3.6)	Visual	04009

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4.3.5.3 Well, fuze (see dwg. 9209333 covering a detail of dwg. 9209330).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:			
101.	Total height	Gage	05001
102.	Diameter of base mating surface (fit with base)	Gage	05002
103.	Diameter of body mating surface (fit with body)	Gage	05003
104.	Profile of upper face	Gage	05004
105.	Wall thickness	Gage	05005
Minor:			
201.	Evidence of poor workmanship (see 3.6)	Visual	05006

4.3.5.4 Well, activator (see dwg. 9209334 covering a detail of dwg. 9209330).

Categories	Defects	Method of Inspection	Code No.
Critical: none defined.			
Major:			
101.	Inside height	Gage	06001
102.	Minor diameter of thread	Gage	06002
103.	Pitch diameter of thread	Gage	06003
104.	Thread length, min	Gage	06004
105.	Inside diameter	Gage	06005
106.	Outside diameter (fit with base and body)	Gage	06006
107.	Flange diameter	Gage	06007
108.	Flange thickness	Gage	06008
109.	Cracks or punctures	Visual	06009
110.	True position of inside diameter with thread	Gage	06010
Minor:			
201.	Evidence of poor workmanship (see 3.6)	Visual	06011

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4.3.5.5 Cover, booster well (see dwg. 9209336 covering a detail of dwg. 9209330).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:			
101.	Diameter	Gage	07001
102.	Thickness	Gage	07002
Minor:			
201.	Evidence of poor workmanship (see 3.6)	Visual	07003

4.3.5.6 Cover, filling hole (see dwg. 9209335 covering a detail of dwg. 9209330).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:			
101.	Diameter	Gage	08001
102.	Thickness	Gage	08002
Minor:			
201.	Evidence of poor workmanship (see 3.6)	Visual	08003

4.3.5.7 Handle, carrying cord (see dwg. 9209337 covering a detail of dwg. 9209330).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major: None defined.			
Minor:			
201.	Inside width	Gage	09001
202.	Outside width	Gage	09002

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203.	Inside height	Gage	09003
204.	Outside height	Gage	09004
205.	Evidence of poor workmanship (see 3.6)	Visual	09005

4.4 Test methods and procedures

4.4.1 Internal air pressure.-The mine assemblies shall be subjected to an internal air pressure, introduced through the filler opening, of two p.s.i.g. for a period of ten seconds, min. Observation shall be made to determine compliance.

4.4.2 Lug pull.-The lugs of the mine assemblies shall be subjected to a pull of 100 pounds, min. The inspector, prior to testing, or as required, shall check the calibration of the test equipment. The accuracy of the load-recording mechanism; i.e., its ability to accurately trace and indicate the time load during application, shall be checked prior to use of the 0.5 inchs per minute maximum travel application rate. If any appreciable difference in reading or result is attributed to rate of travel application, the rate of travel shall be applied at the slowest possible setting. Observation shall be made for any evidence of cracking or separation to determine compliance.

4.4.3 Susceptibility to cracking at cold temperature.-The mine assembly shall be placed in a cold tmperature chamber in such a manner that no external mechanical stresses are applied. The assembly shall be maintained at a temperature of minus 65 degrees Fahrenheit plus or minus 5 degrees for a period of 24 hours, plus or minus 1/2 hour. Observation shall be made for evidence of cracks, including hairline cracks, as evidence of noncompliance.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C.-The assemblies, booster well covers, and filling hole covers for Mine, AT, NM, M19 shall be packaged in accordance with MIL-STD-1169 in such a manner that no damage will occur during shipment.

5.2 Packing

5.2.1 Level C.-The assemblies, booster well covers, and filling hole covers for Mine, AT, NM, M19 shall be packed in

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accordance with MIL-STD-1169. Separately designated items shall not be packaged together.

5.3 Shipping.-When component parts from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated.

5.4 Marking and Labeling.-Marking of cartons and tubes shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. They should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.3).

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of

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inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

Custodian:
Army-MU

Preparing Activity:
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